Ledger core: architecture and trade-offs

Companion to `github.com/mark-per/ledger-core`. Decisions and the risks they defer, arising from the implementation; the rules themselves are in the repository's DESIGN.md and are not repeated here.

The implementation is five domain areas and one orchestrator: **money** (the shared value type at the currency's scale), the **journal** (append-only entries with a booking date and a value date; every balance derived on demand, nothing stored), the **holds book** (a projection beside the journal: promises, never movements), **fees** and **interest** (rules that read closing balances and write entries), and the **replay** (owns the clock, decides nothing about money). One principle drives every decision below: a day's closing balance is a question that can be re-asked later and get a different answer, because entries arrive after the day they belong to.

1. Append-only at scale

Where state grows without bound

Five structures only ever grow: the journal (every entry, forever); the holds book (every hold ever placed, including declined and expired, kept so the report can show them); the interest accruals (one unrounded figure per account per day); the set of processed operation ids that makes delivery idempotent; and the reporter's last-reported closing per account per day. None is trimmed, by design: append-only was the brief's rule and the simplest way to honour it was to keep everything.

What breaks first at 100×

Not memory. A hundred times the brief's stream is a thousand entries; a single process holds millions. What breaks is the scan. `closing(account, day)` walks every entry of the account on each call, and a day close calls it once per value day it re-checks, then again for every accrual it recomputes. The close is therefore entries × days: double the history and each close costs four times as much. At 100× volume with a year of history, the fee sweep after a back-dated entry (which re-checks from that value day to today) is where an operator first notices. The journal's linear growth is a cost; its quadratic re-reading is the failure.

The cheapest structural change

A per-account running balance by value date, maintained on append and owned by the journal. An entry adds its amount to its value day's total and to every later day's; a closing balance becomes a lookup; a back-dated entry costs an update from its value day forward, which is the same span the fee sweep already walks. No rule changes and the journal remains the source of truth; the index is a cache the journal keeps consistent, exactly as it already keeps the per-account close marker. The second change, which bounds the hot path rather than speeding it up, is a period cut-off: once a period is closed and its interest paid, carry its final balance forward as one opening figure and stop re-reading its entries. They stay in the journal (append-only is untouched) but leave the working set. The failing test in the repository is the case this change makes explicit: a correction into a paid period becomes a dated adjustment with a reason, not a silent rescan.

2. Value-dated entries in production

Every entry in this ledger has two dates, and the brief's one back-dated debit is enough to show why that is not a detail: it moves a day that had already been reported, charges fees three days late, and changes what the customer is owed. In a UAE-licensed bank the same two dates touch four surfaces.

- **Customer disclosure.** The Central Bank of the UAE's consumer protection regime requires statements to be accurate and fees and interest to be disclosed and justifiable. A back-dated entry changes a statement a customer has already received and can create a fee or an interest difference for a period they believed was closed. Each restatement is therefore a customer-facing event: a corrected statement, a notice for a late fee, a complaint to answer. The report's "restated" lines in this implementation are the seed of that notice.
- **Regulatory and financial reporting.** Period-end positions reported to the regulator and used for interest income under IFRS 9 are value-dated. An entry valued into a reported period changes a figure that has already left the bank. Banks handle this with a closed-period discipline: anything arriving for a closed period is booked as a dated adjustment in the current period, with the original period referenced, and material restatements are disclosed. This ledger has no closed periods yet; every day is open to back-dating, and section 1 names the fix.

- **Financial-crime monitoring.** Transaction monitoring and reporting under the UAE AML/CFT framework runs on when the bank learned of a transaction, not when it is deemed effective. The booking date is the monitoring date; the value date is a reportable attribute. A ledger that kept only value dates would hide the lag between the two, which is itself a signal.
- **Internal fraud.** The ability to post an entry with an arbitrary past value date is the ability to move interest between customers and periods. Banks treat it as a privileged action: limited to a role, limited in how far back it may reach, and never done by one person alone.

The one control before going live: a back-valuation gate in the journal itself. An entry whose value date is older than a configured limit, or falls in a closed period, cannot be appended directly; it needs a second approver and a reason, it is booked as a dated adjustment referencing the period it corrects, and it emits an event that drives the customer notice and the regulatory restatement log. The gate is one check in `Ledger.append` plus a reason and approver on the entry, both already half present as the `Reason` type on reversals. Everything else in the four surfaces above is a consumer of that event. I would add this before the business-day calendar and before persistence, because it is the only one of the three that stops money from being moved wrongly rather than recording it well.

3. Authorization lifecycle

In this model an authorization is a hold in the holds book, created when the issuer says yes or no and never a journal entry. It ends by a matching settlement (an active hold, a settlement at or below its amount, residual released) or in one of the ways below. Each row states what the model does today and what production would mandate.

Ending	Real-world scenario	Behaviour mandated	In the model
Declined at placement	Available balance after the hold would be below the account's overdraft limit.	No hold counts against the account; the decision is final at that moment and is never re-evaluated when a later back-dated entry changes the balance. The declined hold is kept so a later settlement quoting it is recognised, and so the report shows it.	Yes
Expiry	The merchant never clears (abandoned order, system failure). Schemes release after about 7 days, up to 30 for hotels and car rental.	Released at the close of the expiry day, available balance restored, customer informed on the statement. Expiry by merchant category code in production; a single 7 here.	Yes, 7 days
Settlement above the hold	Tips, fuel, hotel incidentals: the clearing exceeds the authorized amount within scheme tolerance.	Book the full clearing (the obligation exists), consume the hold, flag the excess for review. Production adds a tolerance by merchant category above which the excess is disputed.	Yes, flagged, no cap
Settlement after expiry, or against a declined or already-settled hold	Late presentment; a merchant forcing a declined sale; a duplicate clearing file.	Book as an unmatched settlement with the reason named, leave the hold as it was, and raise the dispute right: the customer did not authorize this charge, or has already paid it. The reason on the entry is what the chargeback is filed from.	Yes
Repeated authorization	Network retry after a timeout: the same request arrives twice.	Idempotent on (account, auth id): return the original decision, place nothing, note the repeat. A retry must never double a hold or flip a decision.	Yes
Void / authorization reversal	Merchant cancels before clearing, in full or in part (fuel pump pre-authorization 500, actual 180).	Release the hold, or reduce it to the remaining amount; nothing in the journal. Requires a reversal message type the brief does not have.	Cut
Incremental authorization	Hotel extends a stay; the merchant raises the hold under the same id.	Re-run the availability check for the increment only; approve or decline the increment; the original hold stands either way.	Cut

Multiple clearings	Split shipment: one authorization, several partial clearings over days.	Each clearing consumes part of the hold; the hold closes when exhausted or when the merchant marks the last one. Needs a remaining-amount on the hold.	Cut; first clearing closes the hold
Account closed or frozen	Sanctions hit, death, court order, customer closure while holds are active.	Freeze: holds stay, settlements still book (the obligation exists), nothing new is approved. Close: holds released after expiry, settlements routed to a suspense account.	Cut

Two related events are not endings of an authorization but belong beside them: a chargeback reverses a settlement weeks later (a REVERSAL of the settlement entry, with the dispute as its reason), and a second identifier, the approval code the issuer generates on a yes, would in production be carried on the hold and required on the clearing as proof of approval. The brief's settlements quote only the network's reference, so the model has one identifier.

4. What was cut and why

Simplification	Why, to stay in scope	Production risk deferred
Single-sided ledger, signed amounts	The brief has one party, the customer. Double-entry pays off when the bank's own accounts appear.	No automatic proof that debits equal credits across the bank; reconciliation to the general ledger is manual.
Balances recomputed by scan, no index, no period cut-off	Correct under back-dating with no extra machinery; six days make the cost invisible.	Quadratic day close and an unbounded working set (section 1).
Interest never restated after capitalization	Nothing is paid before the last day of the brief's window, so the case cannot arise inside it. The entry type exists; the rule does not.	A correction into a paid period leaves interest wrong. The known-failing test demonstrates it; the fix is reverse-and-repost plus a clawback policy nobody has written.
Interest paid as the sum of rounded dailies	Matches the brief's identity literally; each printed daily is its own rounding.	Daily rounding leaks up to half a minor unit per account per day. Printed beside the payment; at scale a treasury would want round-once with an audited allocation.
Fee reversed whenever the day is no longer negative	The rule is a function of the day's closing, in both directions; one rule, no special cases.	A bank that keeps the fee when a late deposit cures the overdraft needs a reason-aware rule; this one gives the fee back regardless of cause.
Fee defined for AED only	The brief gives no BHD fee; inventing one would be a number nobody set.	A BHD overdraft is reported, not charged. Harmless here; a revenue leak if a BHD account were live.
Overdraft limit fixed at zero	The brief's approval rule is the limit-zero special case. The knob exists on the account; nothing sets it.	None until products with overdraft facilities exist; then the limit becomes a credit decision with its own controls.
Hold endings: no void, incremental, multiple clearing, freeze (section 3)	None appears in the stream; each is a message type the brief lacks.	Holds that should have been released stay for 7 days (voids); split shipments over-settle after the first clearing; frozen accounts keep approving.
One identifier per authorization	The brief's settlements quote only the network's reference.	A clearing is matched on the reference alone, with no proof that we approved it. Production adds the issuer-generated approval code on the hold and requires it on the clearing.
Every calendar day is a processing day; value dates are inputs	The brief has no calendar; nothing in the rules needs one.	No weekend or holiday handling for value dates the bank itself assigns; no cut-off time. Both live upstream of this core and in its scheduler.
Day close is a command driven by the stream; single writer, no locks	Keeps the core free of clocks and threads; testable in milliseconds.	Concurrent appends to one account would corrupt sequence order; production needs one writer per account or a lock, and a business-date roll so the close reads a closed set.
No persistence, no recovery	The brief forbids it.	The journal is lost with the process. Production appends each operation to a durable log before the ledger sees it and recovers by replaying that log into a fresh ledger: the holds book and accruals are

		projections, and the stress tests already check that the same stream yields the same report. A bug fixed after the fact is corrected the same way, by replay, never by editing state.
Refusals as report notes, not stored events	Declines and rejections move no money, so they are not journal entries.	No queryable history of what was refused and why. Production routes every refusal and flag (declined authorization, rejected reversal, unmatched settlement, late or duplicate delivery) to a dead-letter topic with its reason, so engineering can fix the cause and re-submit the operation as a new one; nothing is edited.
Operations after the window close normally; no second period	Lets the after-capitalization case exist for the failing test.	Accruals for days after the window are computed and never paid; a real ledger needs a period calendar and a capitalization per period.
Amounts are positive; the kind gives the direction	A debit of 100 is 100 out; "debit of −100" is refused rather than read as 100 in.	Message formats that encode direction in the sign (a refund as a negative debit) must be translated before they reach the ledger.
One currency per account, no conversion	The brief has one currency per account and no exchange rate.	A BHD amount sent to the AED account is refused. No FX and no multi-currency accounts; both are products a bank sells.

The pattern across the table: what was cut is either a message type the brief does not contain or a production concern (persistence, concurrency, calendars, periods) that the brief excluded. The rules themselves are complete for the stream given, and each cut names the one change that would be made first.